

1. Introduction

In this report, I explore passive reconnaissance, a key phase in cybersecurity that involves gathering information about a target without direct interaction thus minimizing the risk of detection. This is different from active reconnaissance, which involves direct engagement with the target and has a higher risk of alerting them. Command-Line Tools for Passive Reconnaissance

1. whois: Queries WHOIS servers for domain registration details, revealing information about domain ownership and contact details.

2. nslookup: Queries DNS servers to obtain DNS records, helping map out the target's DNS infrastructure.

3. dig: Similar to nslookup, but provides more detailed DNS information and flexibility.

Online Services for Passive Reconnaissance

1. DNSDumpster: Collects information about DNS servers, subdomains, and IP addresses associated with a target domain.

2. Shodan.io: Identifies internet-connected devices and systems, revealing exposed services, open ports, and potential vulnerabilities. These tools and services allow us to gather valuable information about our target discreetly

2. Module Sections

Q - You visit the Facebook page of the target company, hoping to get some of their employee names. What kind of reconnaissance activity is this? (A for active, P for passive) Answer is passive as you are not directly engaging with the target.

Q - You ping the IP address of the company webserver to check if ICMP traffic is blocked. What kind of reconnaissance activity is this? (A for active, P for passive) Answer is active as you are directly engaging with the target by sending packets

Q - You happen to meet the IT administrator of the target company at a party. You try to use social engineering to get more information about their systems and network infrastructure. What kind of reconnaissance activity is this? (A for active, P for passive) Answer is active as you are directly engaging with a member of the target company

The screenshot shows a web browser window with the URL `https://tryhackme.com/room/passiverecon`. The browser's address bar and tabs are visible. Below the browser, a notification box says "Woop woop! Your answer is correct". The main content area is titled "Answer the questions below" and contains three questions, each with a text input field and a "Correct Answer" button.

Considering the invasive nature of active reconnaissance, one can quickly get into legal trouble unless one obtains proper authorization.

Answer the questions below

You visit the Facebook page of the target company, hoping to get some of their employee names. What kind of reconnaissance activity is this? (A for active, P for passive)

P ✓ Correct Answer

You ping the IP address of the company webserver to check if ICMP traffic is blocked. What kind of reconnaissance activity is this? (A for active, P for passive)

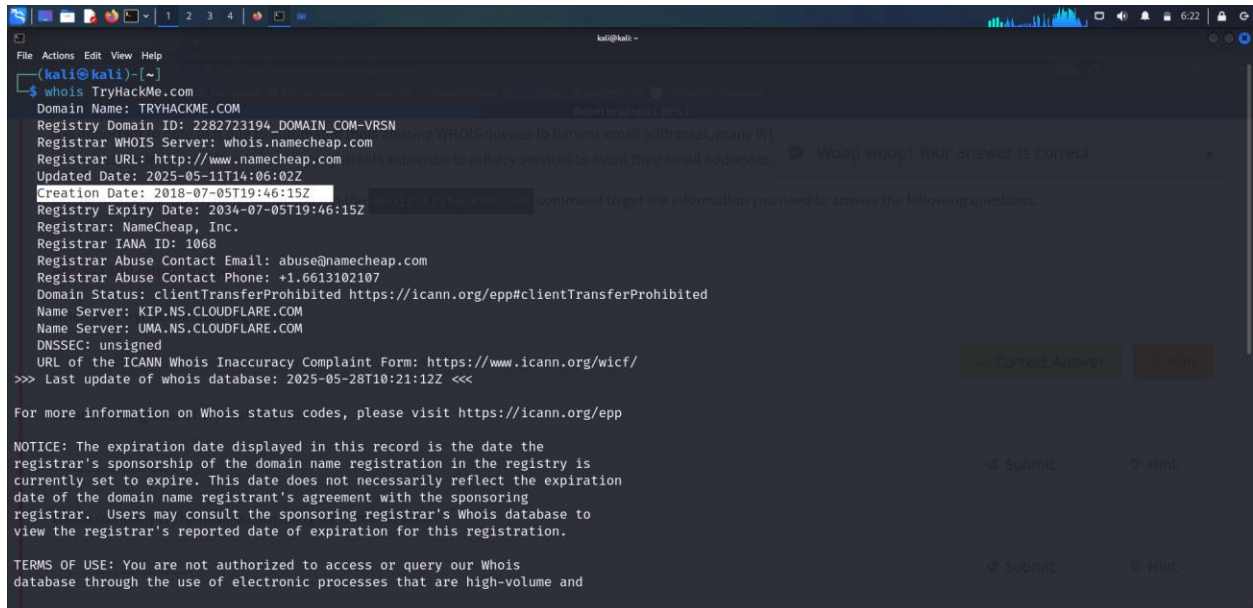
A ✓ Correct Answer

You happen to meet the IT administrator of the target company at a party. You try to use social engineering to get more information about their systems and network infrastructure. What kind of reconnaissance activity is this? (A for active, P for passive)

A ✓ Correct Answer

Q - When was TryHackMe.com registered?

I used the command whois to query the WHOIS records of the specified domain and got the answer as shown in the image below.



```
(kali@kali)-[~]
$ whois TryHackMe.com
Domain Name: TRYHACKME.COM
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-05-11T14:06:02Z
Creation Date: 2018-07-05T19:46:15Z
Registry Expiry Date: 2034-07-05T19:46:15Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: KIP.NS.CLOUDFLARE.COM
Name Server: UMA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-05-28T10:21:12Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
```

Q - What is the registrar of TryHackMe.com? I used the command whois to query the WHOIS records of the specified domain and got the registrar as shown in the image below.

The screenshot shows a Kali Linux terminal window with the following output for the command `whois TryHackMe.com`:

```
(kali@kali)~$ whois TryHackMe.com
Domain Name: TRYHACKME.COM
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-05-11T14:06:02Z
Creation Date: 2018-07-05T19:46:15Z
Registry Expiry Date: 2034-07-05T19:46:15Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: KIP.NS.CLOUDFLARE.COM
Name Server: UMA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-05-28T10:21:12Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

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view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
```

Below the terminal, a web browser window shows the TryHackMe room page for "Passive Recon". A notification box states: "Woop woop! Your answer is correct". The room progress is 46%.

On the AttackBox, open the terminal and run the `whois tryhackme.com` command to get the information you need to answer the following questions.

Answer the questions below

When was TryHackMe.com registered?

20180705

✓ Correct Answer

🔍 Hint

What is the registrar of TryHackMe.com?

namecheap.com

✓ Correct Answer

🔍 Hint

Which company is TryHackMe.com using for name servers?

Submit

🔍 Hint

Task 4 ☐ nslookup and dig

Q - Which company is TryHackMe.com using for name servers? I used the command whois to query the WHOIS records of the specified domain and got the name servers for the specified domain as shown in the image below.

The image shows a web browser window at the URL `https://tryhackme.com/room/passiverecon`. The browser's address bar and tabs are visible. The page content includes a notification that says "Room progress (53%)", a prompt to run the `whois tryhackme.com` command, and a confirmation message "Woop woop! Your answer is correct". Below this, there are three questions with input fields and "Correct Answer" and "Hint" buttons:

- Question: "When was TryHackMe.com registered?"
Answer: `20180705`
- Question: "What is the registrar of TryHackMe.com?"
Answer: `namecheap.com`
- Question: "Which company is TryHackMe.com using for name servers?"
Answer: `cloudflare.com`

Below the questions, there is a "Task 4" section with the text "nslookup and dig".

The bottom part of the image shows a terminal window with the following output:

```
kali@kali:~$ whois TryHackMe.com
Domain Name: TRYHACKME.COM
Registry Domain ID: 2282723194_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.namecheap.com
Registrar URL: http://www.namecheap.com
Updated Date: 2025-05-11T14:06:02Z
Creation Date: 2018-07-05T19:46:15Z
Registry Expiry Date: 2034-07-05T19:46:15Z
Registrar: NameCheap, Inc.
Registrar IANA ID: 1068
Registrar Abuse Contact Email: abuse@namecheap.com
Registrar Abuse Contact Phone: +1.6613102107
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Name Server: KIP.NS.CLOUDFLARE.COM
Name Server: UMA.NS.CLOUDFLARE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-05-28T10:21:12Z <<<

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NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
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registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
```

Q - Check the TXT records of thmlabs.com. What is the flag there? I used the command nslookup and specified the option as TXT to get the text records for the specified domain, thmlabs.com, as shown in the image below and got the flag.

The top screenshot shows a terminal window on Kali Linux. The user runs the command `nslookup -type=txt thmlabs.com`. The output shows a communications error, but the non-authoritative answer for the TXT record is `THM{a5b83929888ed36acb0272971e438d78}`. Below the terminal, a web form asks the user to check the TXT records of thmlabs.com and provides a submit button.

The bottom screenshot shows the TryHackMe web interface. A notification says "Woop woop! Your answer is correct". The page explains that `dig` returns more information than `nslookup` and provides the correct answer: `THM{a5b83929888ed36acb0272971e438d78}`. At the bottom, there are task cards for "Task 5 DNSDumpster" and "Task 6 Shodan.io".

```
(kali@kali)-[~]
$ nslookup -type=txt thmlabs.com
;; communications error to 192.168.120.2#53: timed out
Server:      192.168.120.2
Address:     192.168.120.2#53

Non-authoritative answer:
thmlabs.com  text = "THM{a5b83929888ed36acb0272971e438d78}"

Authoritative answers can be found from:

(kali@kali)-[~]
$
```

Answer the questions below

Check the TXT records of thmlabs.com. What is the flag there?

THM{a5b83929888ed36acb0272971e438d78}

Woop woop! Your answer is correct

A quick comparison between the output of `nslookup` and `dig` shows that `dig` returned more information, such as the TTL (Time To Live) by default. If you want to query a `1.1.1.1` DNS server, you can execute `dig @1.1.1.1 tryhackme.com MX`.

Using the AttackBox, open the terminal and use the `nslookup` or `dig` command to get the information you need to answer the following question.

Answer the questions below

Check the TXT records of thmlabs.com. What is the flag there?

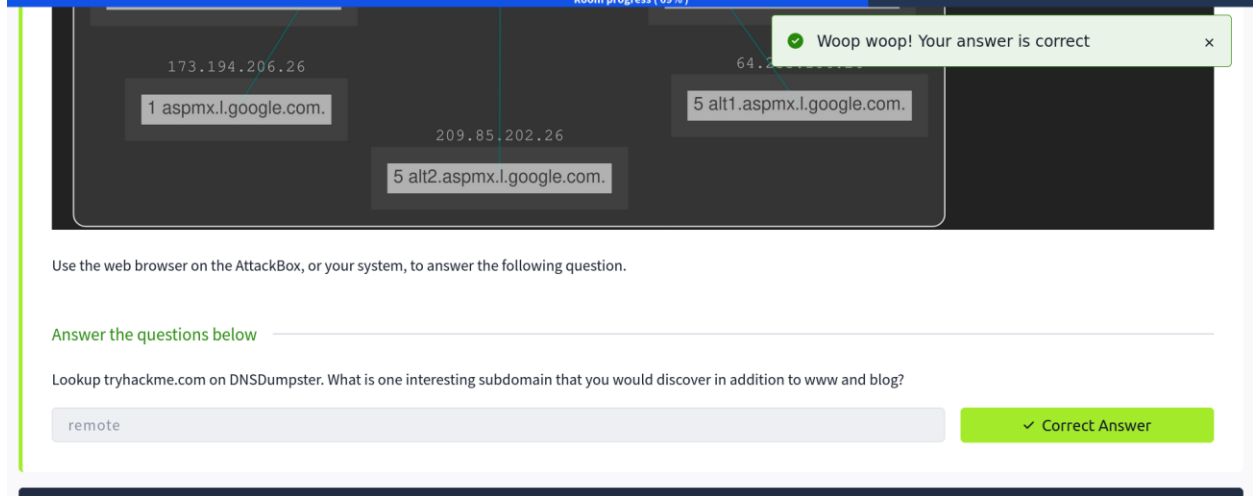
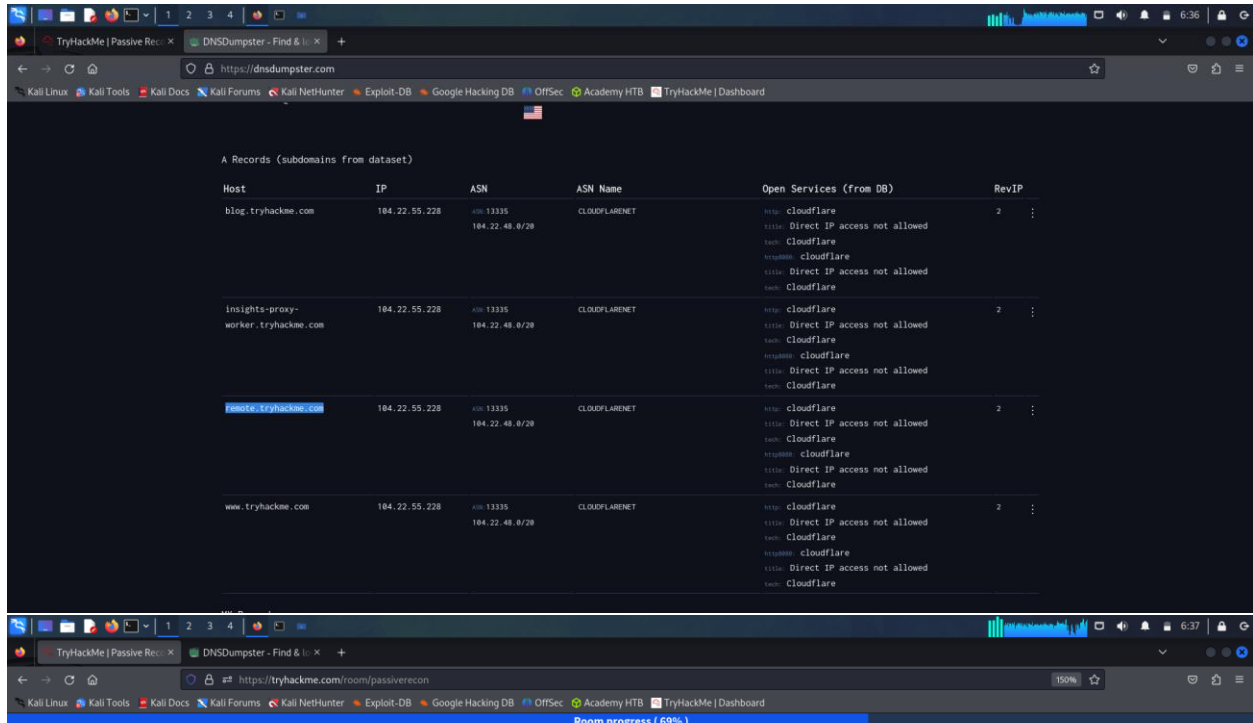
THM{a5b83929888ed36acb0272971e438d78}

Correct Answer

Task 5 DNSDumpster

Task 6 Shodan.io

Q - Lookup tryhackme.com on DNSDumpster. What is one interesting subdomain that you would discover in addition to www and blog? I visited the website <https://dnsdumpster.com/> and searched for tryhackme.com and found 4 subdomains. The interesting one to me was remote.tryhackme.com



Q - According to Shodan.io, what is the 2nd country in the world in terms of the number of publicly accessible Apache servers? I visited the site Shodan.io and searched for Apache and found that China is the 2nd country in the world in terms of the number of publicly accessible Apache servers.

TryHackMe | Passive Recon | Apache - Shodan Search

Room progress (76%)

You may also try searching for the IP addresses you have obtained from DNS lookups. These are, of course, more subject to change. On their [help page](#), you can learn about all the search options available at Shodan.io, and you are encouraged to join TryHackMe's [Shodan.io](#).

Woop woop! Your answer is correct

It would be best to visit Shodan.io to answer the following questions; however, note that you can find the answers on Shodan.io without needing a premium account.

Answer the questions below

According to Shodan.io, what is the 2nd country in the world in terms of the number of publicly accessible Apache servers?

China

✓ Correct Answer

Hint

Based on Shodan.io, what is the 3rd most common port used for Apache?

Submit

Hint

Based on Shodan.io, what is the 3rd most common port used for nginx?

Submit

Hint

Screenshot taken

View image

SHODAN

Explore Pricing Apache

18,032,756

TOP COUNTRIES

United States 4,520,315

China 1,862,743

Germany 1,710,140

Japan 1,532,735

France 697,152

More...

TOP PORTS

80 6,229,033

443 5,242,828

8080 340,516

5006 156,571

8081 145,852

Product Spotlight: We've launched a new API for Fast Vulnerability Lookups. Check out [CVEDB](#)

KACE Systems Management Appliance Service Center

56.155.31.30

HTTP/1.1 200 OK

Date: Wed, 28 May 2025 18:29:37 GMT

Server: Apache

Expires: Wed, 28 May 2025 18:29:37 GMT

Cache-Control: no-store, no-cache, must-revalidate

Pragma: no-cache

Set-Cookie: sbosid=a183c5er31fwup731ey38qag3kndp5; path=/; secure; HttpOnly; SameSite=La...

X-Content-Type-Options: ...

cloud

400 Bad Request

118.99.78.189

HTTP/1.1 400 Bad Request

Date: Wed, 28 May 2025 18:29:18 GMT

Server: Apache/2.4.56 (Ubuntu) OpenSSL/1.1.1f PHP/8.2.4

Content-Length: 875

Connection: close

Content-Type: text/html; charset=iso-8859-1

193.23.249.174

www.sandoc.vip

HTTP/1.1 400 Bad Request

Date: Wed, 28 May 2025 18:29:33 GMT

Server: Apache/2.4.41 (Ubuntu)

Strict-Transport-Security: max-age=3072000; includeSubdomains; preload

Content-Length: 219

Connection: close

Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2...

Q - Based on Shodan.io, what is the 3rd most common port used for Apache? I visited the site Shodan.io and searched for Apache and found that port 8080 is the 3rd most common port used for Apache.

The screenshot shows the Shodan.io search results for the query 'Apache'. The interface includes a sidebar with filters for countries, ports, and organizations. The main area displays search results for Apache servers, including details like IP address, date, server version, and content type. The top results show a high number of bad requests and 400 errors.

| Filter | Value |
|-------------------|--|
| TOTAL RESULTS | 18,032,756 |
| TOP COUNTRIES | United States (4,520,315), China (1,862,743), Germany (1,710,140), Japan (1,532,735), France (697,152) |
| TOP PORTS | 80 (6,229,033), 443 (5,242,828), 8080 (340,516), 5006 (196,571), 8081 (146,952) |
| TOP ORGANIZATIONS | China Education and Research Network (901,443) |

It would be best to visit Shodan.io to answer the following questions; however, note that you can find the answers on Shodan.io without needing a premium account.

Woop woop! Your answer is correct

Answer the questions below

According to Shodan.io, what is the 2nd country in the world in terms of the number of publicly accessible Apache servers?

China

Correct Answer

Hint

Based on Shodan.io, what is the 3rd most common port used for Apache?

8080

Correct Answer

Hint

Based on Shodan.io, what is the 3rd most common port used for nginx?

Submit

Hint

Task 7 Summary

Q - Based on Shodan.io, what is the 3rd most common port used for nginx? I visited the site Shodan.io and searched for nginx and found that port 5001 is the 3rd most common port used for nginx.

The top screenshot shows the Shodan.io search results for the query 'nginx'. The total results are 57,197,068. The top countries are China (14,011,784), United States (10,293,867), Hong Kong (4,064,153), Japan (3,305,180), and Germany (3,212,166). The top ports are 80 (10,647,662), 443 (8,434,776), 5001 (688,401), 5000 (620,583), and 888 (526,868). The top organizations are Metaverse Limited (4,210,452). The bottom screenshot shows a TryHackMe room interface with a progress bar at 92% and a notification that the answer is correct.

It would be best to visit Shodan.io to answer the following questions; however, note that you can find the answers on Shodan.io without needing a premium account.

✓ Woop woop! Your answer is correct

Answer the questions below

According to Shodan.io, what is the 2nd country in the world in terms of the number of publicly accessible Apache servers?

China

✓ Correct Answer

🔍 Hint

Based on Shodan.io, what is the 3rd most common port used for Apache?

8080

✓ Correct Answer

🔍 Hint

Based on Shodan.io, what is the 3rd most common port used for nginx?

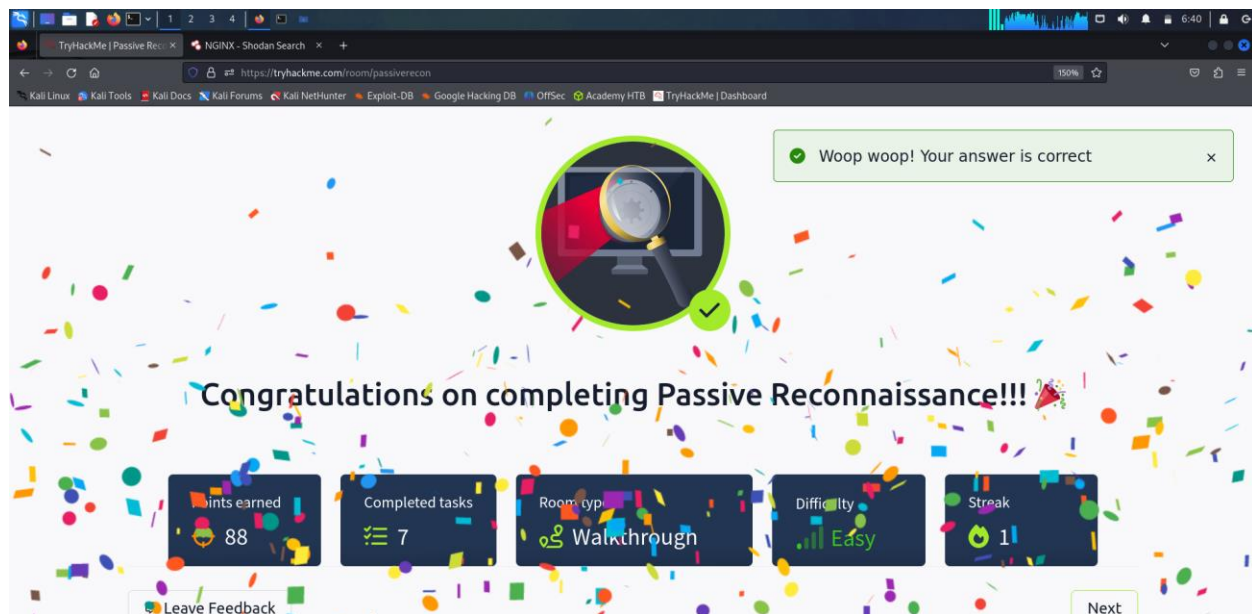
5001

✓ Correct Answer

🔍 Hint

Task 7 Summary

3. Shareable Link and Final Screenshot



4. Social Media Posting

https://x.com/intent/post?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fpassiverecon%3Futm_source%3Dtwitter%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&text=Share%20your%20achievement&via=realtryhackme&hashtags=tryhackme

https://www.facebook.com/sharer/sharer.php?u=https%3A%2F%2Ftryhackme.com%2Froom%2Fpassiverecon%3Futm_source%3Dfacebook%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom&t=Share%20your%20achievement

https://www.linkedin.com/sharing/share-offsite/?url=https%3A%2F%2Ftryhackme.com%2Froom%2Fpassiverecon%3Futm_source%3Dlinkedin%26utm_medium%3Dsocial%26utm_campaign%3Dsocial_share%26utm_content%3Droom

5. Conclusion

In this module on passive reconnaissance, I explored essential tools and services for discreet information gathering in cybersecurity. I understood how to use the following command line tools: whois: Reveals domain registration details. nslookup: Queries DNS servers for various records. dig: Provides detailed DNS information. Additionally, I used the following online services: DNSDumpster: Gathers comprehensive DNS information. Shodan.io: Identifies internet-connected devices and potential vulnerabilities. This module highlighted the importance of discretion and the vast publicly available information that can be harnessed. Proficiency in these tools is vital for future cybersecurity professionals, aiding in safeguarding digital assets and anticipating threats.